



Chapter 8- Device Drivers & Device Manager



1. What is a Driver? Why is it Important?

A **device driver** is a special type of software that helps the **Operating System (OS)** communicate with **hardware devices**.



Example:

- When you connect a **printer to a USB port**, the **motherboard cannot automatically understand** what device is connected.
- The **driver identifies the device** and tells the OS:
 - What the device is (printer, keyboard, etc.)
 - How to communicate with it



Without a driver:

- The device may not work at all
- Or may show as **“Unknown Device”**



Simple real-life analogy:

Driver = Translator between **human (OS)** and **machine (hardware)** 



2. Drivers Exist for All Components

Not just external devices — even internal components need drivers:

- BIOS
- Chipset
- I/O Chip
- Processor (CPU)
- GPU
- Motherboard components



These drivers ensure **smooth communication between hardware and OS**

! 3. Keyboard Not Working After Lock/Unlock (Real Scenario)

Problem:

- System is ON
- You lock → unlock
- Keyboard stops working

Error Seen:

“Windows cannot start this hardware device because its configuration information in the registry is incomplete or damaged (Code 19)”

Temporary Solution:

1. Open **Device Manager**
2. Right-click keyboard driver
3. Click **Uninstall Device**
4. Refresh / Scan for hardware changes

 This reloads the driver and keyboard starts working again

Permanent Solution (Important for Interviews):

Fix Registry Corruption (Code 19):

1. Press Win + R → type regedit
2. Navigate to:

HKEY_LOCAL_MACHINE
SYSTEM
CurrentControlSet

Control Class

3. Find keyboard class GUID:

{4D36E96B-E325-11CE-BFC1-08002BE10318}

4. Delete:

- UpperFilters
- LowerFilters

5. Restart system



Other Permanent Fixes:

- Update keyboard driver from official website
- Run Windows Troubleshooter
- Check for Windows updates
- Disable Fast Startup
- Check USB port / try different keyboard



4. Device Manager – Complete Breakdown

Device Manager stores **all connected device information**



Categories Explained:



1. Audio Inputs and Outputs

- Shows speakers, headphones, Bluetooth audio
- Used to select output device



Example: Switching from speaker to Bluetooth earbuds

2. Audio Processing Objects

- Enhances sound (noise cancellation, effects)
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3. Batteries

- Shows battery status
 - Without it → no battery percentage
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4. Biometrics

- Fingerprint & Face recognition
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5. Bluetooth

- Shows all connected/past Bluetooth devices
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6. Disk Drives

- Hard disk / SSD
-

7. Display Adapters (VERY IMPORTANT)

 Key point:

- CPU = processing
- GPU = how things are displayed (graphics, animation)

 If system is slow:

- Update **graphics driver**

 Do NOT rely on Device Manager update

 Always download from official website

8. Firmware

- BIOS related drivers
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9. Human Interface Devices (HID)

- Keyboard, mouse, touch devices
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10. IDE ATA / SATA Controllers

- Storage communication
-

11. Imaging Devices

- Camera
-

12. Keyboards

- Built-in + external
-

13. Mice

- All pointing devices
-

14. Monitors

- Display screens
-

15. Network Adapters

Includes:

- WiFi
- Ethernet
- Bluetooth

- Virtual adapters (VMware etc.)
-

16. Other Devices

- Drivers not installed yet
-

17. Printers

18. Security Devices

- TPM (Trusted Platform Module)

 Required for Windows 11 installation

19. System Devices

- Motherboard chips
 - Intel chipset, serial IO, etc.
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20. USB Controllers

- All USB ports and connected devices
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5. How to Download Drivers (Correct Method)

 Don't trust Device Manager updates

 Always use **official website**

Steps:

1. Search:

HP driver download / Dell driver download

2. Enter:

- Serial Number / Product Number / Model Name
-

To Find Serial Number:

Run command:

wmic bios get serialnumber

6. Important Drivers to Update

Must Update:

- BIOS (System BIOS)
 - Graphics driver (Display adapter)
 - Chipset driver
 - Intel Management Engine
 - Network drivers:
 - Ethernet
 - WiFi (Realtek / Intel)
 - Bluetooth
 - Audio driver (Realtek HD Audio)
 - TPM / Security drivers
-

Performance Impact:

Problem	Driver to Check
System slow	Graphics + Chipset
Network slow	Network adapter
Audio issue	Audio driver
Camera not working	Camera driver
Storage issue	Disk driver

7. Troubleshooting WiFi Slow Issue (Interview Scenario)

Step 1: Ask Question

 Is issue:

- Only for one user? → System issue
 - For everyone? → Network issue
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Step 2: If Single User

1. Open Device Manager
 2. Go to Network Adapter
 3. Check driver (Intel Wireless / Realtek)
 4. Update from official website
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8. Common Device Manager Error Codes

These are VERY IMPORTANT for interviews:

What to Do?

-  Double-click device
 -  Check **Device Status**
-

Common Error Codes + Solutions

Code 10 – Device Cannot Start

Cause:

- Driver issue
- Hardware failure

Solution:

- Update driver
 - Reinstall driver
 - Check hardware
-

Code 19 – Registry Corrupted (VERY IMPORTANT)

Error:

"Configuration information in registry is incomplete or damaged"

Temporary Fix:

- Uninstall driver
 - Refresh
-

✅ **Permanent Fix:**

👉 **Steps:**

1. Open **Registry Editor** (regedit)

2. Go to:

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Class

3. Delete:

- UpperFilters
- LowerFilters

4. Restart system

🔴 **Code 25 – Device Not Working Properly**

👉 **Solution:**

- Restart system
 - Update driver
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🔴 **Code 28 – Driver Not Installed**

👉 **Solution:**

- Install driver manually
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🔴 **Code 32 – Driver Disabled**

👉 **Solution:**

- Enable driver
 - Reinstall
-

● Code 39 – Driver Corrupted

👉 Solution:

- Reinstall driver
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● Code 52 – Digital Signature Issue (Cyber Security 🔥)

👉 Cause:

- Unsigned driver
 - Malware-modified driver
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⚠️ **Solution:**

- Install signed driver
 - Disable driver signature enforcement (temporary)
 - Scan for malware
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🔧 **General Fix Approach:**

1. Check error code
 2. Reinstall driver
 3. Update driver (official site)
 4. Check hardware connection
 5. Run troubleshooter
 6. Check registry (if needed)
 7. Restart system
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Final IT Support Tip

 For fast troubleshooting:

- Always identify **root cause first**
 - Don't randomly try solutions
 - Follow structured approach:
 1. Identify problem
 2. Ask questions
 3. Check Device Manager
 4. Update/reinstall driver
 5. Test
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Quick Summary

- Drivers = bridge between OS & hardware
- Device Manager = all device control center
- Always use official drivers
- Learn error codes (very important)
- Troubleshooting = logical + step-by-step

Chapter 8 (SUMMARY)

What is Driver?

 Driver = Software that connects OS with hardware

 Example:

- Printer → needs driver
- WiFi → needs driver

 Without driver:

- ✗ Device won't work
- ✗ Shows "Unknown Device"

Where Drivers are Used?

 All hardware needs drivers:

- CPU
- GPU
- Chipset
- BIOS
- Network

Device Manager

 Central tool to manage all devices

 Open:

 Right-click Start → Device Manager

Important Sections

- Display Adapter 
- Network Adapter 
- Disk Drives 
- HID (Keyboard/Mouse) 
- USB Controllers 

Important Concept

-  If system slow:
-  Check **Graphics driver + Chipset driver**

Driver Download Rule

-  Don't trust auto update
-  Always use official website

Important Error Codes

Code	Meaning	Fix
10	Device cannot start	Reinstall driver
19	Registry corrupted	Fix registry
28	Driver not installed	Install driver
32	Driver disabled	Enable
39	Driver corrupted	Reinstall
52	Signature issue	Install signed driver

Real Troubleshooting Logic

 Always:

1. Identify problem
 2. Check Device Manager
 3. Update/reinstall driver
 4. Test
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CONCLUSION

 Key learning:

- Drivers = Communication bridge 
- Device Manager = Control center 
- Error codes = Diagnosis tool 

 Final line:

“Good IT Support engineer doesn’t guess, he checks Device Manager first”



Q & A

◆ DRIVER QUESTIONS

? Q1: What is a device driver?

👉 Answer:

A device driver is a software that allows the operating system to communicate with hardware devices.

Without a driver, the OS cannot understand how to use the hardware.

? Q2: Why drivers are important?

👉 Answer:

Drivers are important because they act as a translator between hardware and software.

Without drivers, devices either won't work or will work incorrectly.

? Q3: What happens if driver is missing?

👉 Answer:

- Device shows as "Unknown Device"
 - Device will not function properly
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◆ DEVICE MANAGER QUESTIONS

? Q4: What is Device Manager?

👉 Answer:

Device Manager is a Windows tool used to view, manage, and troubleshoot hardware devices and their drivers.

? Q5: How do you open Device Manager?

👉 Answer:

- Right-click Start → Device Manager
 - Or Run → devmgmt.msc
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? Q6: What is Display Adapter?

👉 Answer:

Display Adapter represents the GPU which handles graphics and display output.

? Q7: What is Network Adapter?

👉 Answer:

It represents network devices like WiFi, Ethernet, and Bluetooth.

◆ DRIVER UPDATE QUESTIONS

? Q8: How do you update drivers properly?

👉 Answer:

1. Go to official website (HP/Dell/Lenovo)
 2. Enter model or serial number
 3. Download correct driver
 4. Install manually
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? Q9: Why not use Device Manager auto update?

👉 Answer:

Because it may install outdated or generic drivers, not optimized ones.

◆ ERROR CODE QUESTIONS (VERY IMPORTANT)

? Q10: What is Code 10 error?

👉 Answer:

Device cannot start due to driver or hardware issue.

Fix: Reinstall driver or check hardware.

? Q11: What is Code 19 error?

👉 Answer:

Registry configuration is corrupted

✅ Fix:

- Delete UpperFilters & LowerFilters
 - Restart system
-

? Q12: What is Code 28?

👉 Answer:

Driver not installed.

👉 Install driver manually.

? Q13: What is Code 52?

👉 Answer:

Driver signature issue (security risk).

👉 Fix:

- Install signed driver
 - Scan system
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◆ TROUBLESHOOTING QUESTIONS

KEYBOARD ISSUE

? **Q14: Keyboard not working after unlock?**

👉 **Answer:**

- Go to Device Manager
 - Uninstall keyboard driver
 - Scan for hardware changes
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WIFI SLOW

? **Q15: WiFi slow, how will you troubleshoot?**

👉 **Answer:**

1. Ask if issue is for all users
 2. If single user:
 - Check network adapter driver
 - Update driver
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SYSTEM SLOW

? **Q16: System slow even after OS install?**

👉 **Answer:**

- Check graphics driver
- Check chipset driver
- Update both

CAMERA NOT WORKING

? Q17: Camera not working?

👉 Answer:

- Check Imaging Devices
 - Update camera driver
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AUDIO ISSUE

? Q18: No sound?

👉 Answer:

- Check audio driver
 - Reinstall driver
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◆ REAL JOB SCENARIOS

? Q19: Unknown device in Device Manager?

👉 Answer:

- 👉 Driver missing
 - 👉 Install correct driver from official site
-

? Q20: Device showing yellow exclamation mark?

👉 Answer:

- 👉 Driver issue
 - 👉 Update or reinstall driver
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? Q21: USB device not detected?

👉 Answer:

- Check USB controllers
 - Update driver
 - Try different port
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◆ INTERVIEW IMPRESSION QUESTIONS

? Q22: What is your driver troubleshooting approach?

👉 Answer:

1. Identify device
 2. Check Device Manager
 3. Check error code
 4. Reinstall/update driver
 5. Test
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? Q23: Which drivers are most important?

👉 Answer:

- BIOS ,Chipset ,Graphics ,Network ,Audio